جامعة البلقاء التطبيقية

Al-Balqa Applied University

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Q1- Calculate ΔH for the following reaction

$NO(g) + O(g) \rightarrow NO_2(g)$

Use the following data in your calculation

$NO_{(g)} + O_{3(g)} \rightarrow NO_{2(g)} + O_{2(g)}$	$\Delta H_1 = -198.9$ kJ
$O_{3(g)} \rightarrow 3/2\ O_{2(g)}$	$\Delta H_2 = -142.3$ kJ
$O_{2(g)} \rightarrow 2\ O_{(g)}$	$\Delta H_3 = +495.0$ kJ

A) 304.1 B) -304.1 C) -551.6 D) 551.6

Q2- A 500.0 gram sample of aluminum is initially at 25.0 °C. It absorbs 32.60 kJ of heat from its surroundings. What is its final temperature, in °C? (specific heat = 0.9930 J g 1 °C-1 for aluminum)

A) 40.4 °C B) 64.7 °C C) 65.7 °C D) 90.7 °C

Q3- The type of calorimeter used for determination of enthalpy of formation of magnesium oxide is:

A) Cup-calorimeter of constant volume B) Cup-calorimeter of constant pressure

C) Bomb –calorimeter of constant pressure D) Bomb –calorimeter of constant volume

Q4- A10.0 mL bleach sample was diluted to 100 mL in a volumetric flask. A 25.0 mL of this solution required 12.5 mL of 0.30 M $Na_2S_2O_3$ to reach the endpoint of the titration. Calculate the mass percent of NaOCl in the original sample (Molar mass NaOCl= 74.5 g/mol). Assume the density of bleach solution is 1.084g/mL.

A) 5.15% B) 6.46% C) 0.515% D) 0.646%

Q5- Which of the following statement is correct concerning the titration of I_2 which used in bleach analysis?

A) The indicator must be added at the beginning of the titration.

B) The indicator must be added when the color of I_2 solution turns to dark blue

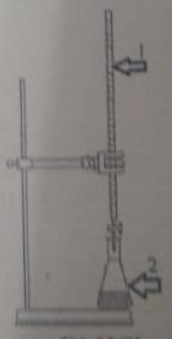
C) The indicator must be added when the color of I_2 solution turns to colorless.

D) The indicator must be added when the color of I_2 solution turns to pale yellow.

Q6- The active ingredient in bleach is:

A) H^+ B) OCl^- C) CH_3COOH D) I^-

Q7- The setup shown on the right is the one you used during analysis of the bleach solution. The name of equipment 1 is---------- and the name of equipment 2 is ---------and the solution filled in equipment 1 is---------- whereas the solution placed inside equipment 2 contain ----------.(The answers respectively are):



A) Burette, flask, bleach with KI and HCL, $Na_2S_2O_3$,

B) Burette, flask, $Na_2S_2O_3$, bleach with KI and HCL

C) Flask, Burette, $Na_2S_2O_3$, bleach with KI and HCL

D) Flask, Burette, bleach with KI and HCL, $Na_2S_2O_3$,

Q8-The name of indicator used in bleach analysis :

A) Phenolnapthlien B) KI C) starch D) HCl

Q9- Which of the following statement is correct concerning acid-base titration experiment :

A) During the titration of vinegar solution the inside walls of titration flask shouldn't rinsed with distilled water

B) The burette should be rinsed with vinegar solution before titration

C) The burette should be rinsed with NaOH solution before titration

D) The indicator in acid-base experiment should be added during the titration

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Q10- A 20.0 ml of NaOH solution was needed to titrate 10.0 ml of 0.5 M acetic acid solution (molar mass = 60.0 g/ mol , D.F= 10) .Calculate the W\W % for the acetic acid in the original vinegar sample (Density of vinegar =1.04 g\mL) .

$$CH_3COOH + NaOH \longrightarrow H_2O + CH_3COO^-Na^+$$

A) 16.7 % B) 11.1 % C) 26.8 % D) 28.9%

Q11-The indicator used during vinegar analysis has a ---------- color in the acidic medium:
A) Yellow B) Colorless C)Pink D) Blue

Q12- The active ingredient in vinegar is :
A) CH_3COOH B) NaOH C) HCl D) H_2O

During limiting reactant experiment, when 30.0 mL of 0.55 M Na_2CO_3 (molar mass 105.99 g/mol) solution were mixed with 15.0 mL of 1.4 M $CaCl_2$ (111.0 g/mol) solution then 1.265 g of $CaCO_3$ (100.09) were obtained. Answer Q13, 14& 15

Q13- The percentage yield of this reaction equal:
A) 45.9 % B) 76.6% C) 46.3 % D) 18.3%

Q14- At the end of reaction, ... moles of ..... is remained unreacted.
A) 0.0015 mole, $CaCl_2$ B) 0.00375 mole, $CaCl_2$ C) 0.0045 mole,$CaCl_2$ D) 0.0015 mole, Na_2CO_3

Q15- The solid formed at the end of the reaction is :
A) Filtered then washed with cold water and then allowed to dry .
B) Washed with cold water then Filtered and then allowed to dry .
C) Allowed to dry then Washed with cold water and then Filtered
D) All of the above

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Q16- The steps (from 1 to 4) written below are the steps followed during the experiment of reaction of $CaCl_2$ solution with Na_2CO_3 solution (limiting reactant experiment) . Arrange the steps correctly according to the experimental procedure performed in your lab.

1. Clean Erlenmeyer flasks were used for mixing the 20 mL of $CaCl_2$ with 10 mL of Na_2CO_3.
2. Two graduated cylinder were used; the first was filled with $CaCl_2$ solution. The second filled with Na_2CO_3 .
3. Filtration was performed to collect the precipitate.
4. After drying the precipitate, it was weighed and calculations were then performed.

A) 1,2,3,4 B) 2,1 ,3,4 C) 2,3,1,4 D) 4,3,2,1

Q17- The following data were collected during the determination of the molecular weight of a volatile liquid. The mass of the condensed vapour of this volatile liquid equals: (R=0.08206L.atm/K.mol, 1atm=760mmHg)

A) 2.8
B) 2.5
C) 1.25
D) 3.5

Molecular Mass of volatile liquid	485.9 g/mol
The temperature of the Boiling water bath	86 °C
Atmospheric pressure	720 mmHg
Total volume of 50 mL flask used to vaporize the unknown volatile liquid	80 mL

Q18-The method you worked out in your lab to determine the molecular weight of volatile liquid is called:

A) Dumas
B) Titration
C) Filtration
D) None of above

Q19- For the hydrated salt: $MgSO_4 . \times H_2O$, if the mass of hydrated salt was 2.0 g and % H_2O = 47.3 %, then the value of (×) is (At.wts (g/mol) Mg:24.3 ,S:32 , O:16)

A) 4 B)6 C)8 D)3

Q20- The general formula of an alum is:

A) $M^{+1}M^{+3}(SO_4)_2.xH_2O$ B) $M^{+2}M^{+2}(SO_4)_2.xH_2O$ C) $M^{+3}(SO_4)_2.xH_2O$ D) $M^{+1}(SO_4)_2.xH_2O$

Q21- If the empirical formula of compound is CH_2O and the molar mass of the compound equals 60 g/m the molecular formula of this compound is:

A) $C_3H_6O_3$ B) CH_2O C) $C_2H_4O_2$ D) $C_6H_{12}O_6$

Q22- The formula of the by product formed during the burning of magnesium turnings is :

A) MgO B) Mg_3N_2 C) Mg_2N_3 D)NH_3

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Q23-Determine the empirical formula of a compound that contain 6.6% H,40.1 %C and 53.3% O : (At.wts g/mol: C=12,O=16,H=1)

A) CH_2O B)C_3H_6 C) C_3H_6O D) C_2H_6O

Q24- The presence of the following notation (shown beside) on a reagent bottle, indicates that this chemical substance is:



A) Flammable B) oxidizing C) toxic D) corrosive

Q25- Only one of the following is considered as a correct laboratory safety rule?

A) You can smell any chemical in your laboratory by holding your face directly over the bottle.

B) Flammable liquid are heated using water bath

C) Clean pipets and droppers are allowed to be inserted in any reagent bottle

D) The fume hood is an equipment used to wash the glassware equipment's from any sticking materials

Q26- Higher pressure ----------the boiling point of a liquid.

A) Increases B) decreases C) does not effect

BEST WISHES